

1. Given a class $\mathcal{F}$ of functions $f : \mathcal{X} \rightarrow \mathbb{R}$, let $\mathcal{H} = \{\text{sgn} \circ f : f \in \mathcal{F}\}$. Let $D = (X_i, Y_i)_{i=1}^n$ be n i.i.d. input-output pairs taking values in $\mathcal{X} \times \{-1, 1\}$. Show that for any $\hat{h} = \text{sgn} \circ \hat{f} \in \mathcal{H}$ depending on D (e.g. the ERM over $\mathcal{H}$), and any $\rho > 0$, we have that the misclassification risk $R(\hat{h})$ satisfies

$$\mathbb{E}R(\hat{h}) \leq \mathbb{E} \left(\frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{\hat{f}(X_i)Y_i \leq \rho\}} \right) + \frac{2}{\rho} \mathcal{R}_n(\mathcal{F}).$$

[Hint: Construct an appropriate surrogate loss ϕ such that $\mathbb{1}_{(-\infty, 0]} \leq \phi \leq \mathbb{1}_{(-\infty, \rho]}$, and ϕ has Lipschitz constant $1/\rho$.]

2. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a strictly convex function and suppose $C \subseteq \mathbb{R}^d$ is a convex set. Suppose $x_1, x_2 \in C$ satisfy $f(x_1) = f(x_2) = \inf_{x \in C} f(x)$. Show that $x_1 = x_2$.
3. Show that for a closed convex set $C \subseteq \mathbb{R}^d$, $\pi \in C$ is a projection of $x \in \mathbb{R}^d$ onto C if

$$(x - \pi)^\top (z - \pi) \leq 0 \quad \text{for all } z \in C.$$

4. Show that $\partial \|\beta\|_1 = \{b : \text{for each } j, b_j \in [-1, 1] \text{ and } b_j = \text{sgn}(\beta_j) \text{ if } \beta_j \neq 0\}$.
5. Show that $x \in \mathbb{R}^d$ minimises $f : \mathbb{R}^d \rightarrow \mathbb{R}$ if and only if $0 \in \partial f(x)$.
6. This question derives the form of the projection onto an ℓ_1 -norm constraint set.
 - (a) Fix $x \in \mathbb{R}^p$ and $\gamma > 0$, and let $g(\beta) := \|\beta - x\|_2^2/2 + \gamma \|\beta\|_1$. Show that g is minimised over $\beta \in \mathbb{R}^p$ by

$$\beta_j^* = \max(|x_j| - \gamma, 0) \text{sgn}(x_j).$$

[Hint: Use 4 and 5.]

- (b) Argue that if β^* above has $\|\beta^*\|_1 = \lambda$, then β^* is the projection of x onto the set $C = \{z : \|z\|_1 \leq \lambda\}$ i.e. $\beta^* = \pi_C(x)$.
7. Consider a version of stochastic gradient descent for minimising

$$f(\beta) = \frac{1}{n} \sum_{i=1}^n \ell(h_\beta(x_i), y_i)$$

(assumed here to be differentiable) over $\beta \in C \subseteq \mathbb{R}^p$ where C is closed and convex, and let $\hat{\beta}$ be the minimiser. We take $U_1, \dots, U_{k-1}$ uniformly distributed on $\{1, \dots, p\}$ and writing $\beta^{(s)} \in \mathbb{R}^p$ for the s th iterate we take

$$\tilde{g}_s = e_{U_s} \frac{\partial f}{\partial \beta_{U_s}} \Big|_{\beta^{(s)}}.$$

Show that under the setup of Theorem 20 on the convergence of gradient descent (so in particular we assume $\sup_{\beta \in C} \|\nabla f(\beta)\|_2 \leq L$, the output $\bar{\beta}$ of the algorithm set out above, with a suitable step size $\eta > 0$ you should specify, satisfies

$$\mathbb{E}f(\bar{\beta}) - f(\hat{\beta}) \leq 2LR\sqrt{\frac{p}{k}}.$$

8. The following result shows that the theory for stochastic gradient descent can be used to obtain some forms of generalisation error bounds. Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. input–output pairs. Consider empirical risk minimisation with logistic loss ϕ where $\mathcal{X} = \{x \in \mathbb{R}^p : \|x\|_2 \leq C\}$ and $\mathcal{H} = \{x \mapsto x^\top \beta : \|\beta\|_2 \leq \lambda\}$. Let π denote projection onto $\{\beta : \|\beta\|_2 \leq \lambda\}$. Let $\beta_1 \in \mathbb{R}^p$ be the 0 vector and define iteratively for $i = 1, \dots, n-1$,

$$g_i = Y_i X_i \phi'(Y_i X_i^\top \beta_i),$$

$$\beta_{i+1} = \pi(\beta_i - \eta g_i).$$

[Note the β_i above are vectors.] Let $\bar{\beta} = \frac{1}{n} \sum_{i=1}^n \beta_i$ and set $\bar{h}(x) = x^\top \bar{\beta}$. Show that for some step size $\eta > 0$ you should specify,

$$\mathbb{E}R_\phi(\bar{h}) - R_\phi(h^*) \leq \frac{2C\lambda}{\log(2)\sqrt{n}}.$$

[Hint: Write the risk itself in the form $\mathbb{E}f(\beta; U)$ for some U .]

9. Consider the Adaboost algorithm with base class $\mathcal{B}$ (satisfying that if $h \in \mathcal{B}$ then $-h \in \mathcal{B}$) and assume that at no iteration does any $h \in \mathcal{B}$ perfectly classify the data.

(a) Show that

$$\frac{\sum_{i=1}^n w_i^{(m+1)}}{\sum_{i=1}^n w_i^{(m)}} = 2\sqrt{\widehat{\text{err}}_m(1 - \widehat{\text{err}}_m)}$$

where $\widehat{\text{err}}_m := \text{err}_m(\hat{h}_m)$.

- (b) Assume that for each m , $\widehat{\text{err}}_m \leq 1/2 - \gamma$ for some $\gamma > 0$. Show that the empirical risk of the Adaboost output decreases exponentially fast with M :

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \exp(-Y_i \hat{f}_M(X_i)) &= \prod_{m=1}^M 2\sqrt{\widehat{\text{err}}_m(1 - \widehat{\text{err}}_m)} \\ &\leq \exp(-2\gamma^2 M). \end{aligned} \tag{1}$$

(c) Let

$$\mathcal{H} = \left\{ \sum_{m=1}^M \beta_m h_m : \|\beta\|_1 \leq 1, h_m \in \mathcal{B} \text{ for } m = 1, \dots, M \right\}.$$

Explain why for $x_{1:n} \in \mathcal{X}^n$,

$$\hat{\mathcal{R}}(\mathcal{H}(x_{1:n})) \leq \sqrt{\frac{2\text{VC}(\mathcal{B}) \log(n+1)}{n}}.$$

- (d) Given input–output pairs $(X_i, Y_i)_{i=1}^n$ taking values in $\mathcal{X} \times \{-1, 1\}$, let $\hat{f}_M = \sum_{m=1}^M \hat{\beta}_m \hat{h}_m$ be the output of the Adaboost algorithm with base classifier class $\mathcal{B}$. With the assumption of (b) that $0 < \widehat{\text{err}}_m \leq 1/2 - \gamma$ for some $1/2 > \gamma > \rho > 0$, show that

$$\frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{\hat{f}_M(X_i) Y_i \leq \rho \|\hat{\beta}\|_1\}} \leq \exp\{-2M(\gamma^2 - c\rho)\}, \quad \text{where } c = \frac{1}{4} \log\left(\frac{1+2\gamma}{1-2\gamma}\right),$$

and $\hat{\beta} := (\hat{\beta}_m)_{m=1}^M$. [Hint: Use $\mathbb{1}_{\{u \leq b\}} \leq \exp(b - u)$ and (1). You may further use the fact that $u \mapsto u^{1-\rho}(1-u)^{1+\rho}$ is increasing for $0 < u < 1/2 - \rho$.]

(e) Show that writing $\hat{h} := \text{sgn} \circ \hat{f}_M$, we have that the misclassification risk $R(\hat{h})$ satisfies

$$\mathbb{E}R(\hat{h}) \leq \exp\{-2M(\gamma^2 - c\rho)\} + \frac{2}{\rho} \sqrt{\frac{2\text{VC}(\mathcal{B}) \log(n+1)}{n}}.$$

[Hint: Recall Qu. 1.]

10. Let $\mathcal{X} = \mathbb{R}^p$. Consider performing gradient boosting with base regression procedure the empirical risk minimiser over $\mathcal{H} = \{x \mapsto \mu + x_j\beta : j \in \{1, \dots, p\}, \mu, \beta \in \mathbb{R}\}$ with squared error loss. Consider the m th iteration. Show that $\hat{g}_m(x) = \hat{\mu}_{\hat{j}} + x_{\hat{j}}\hat{\beta}_{\hat{j}}$ where

$$\hat{\beta}_j := \frac{\sum_{i=1}^n (W_i - \bar{W})(X_{ij} - \bar{X}_j)}{\sum_{i=1}^n (X_{ij} - \bar{X}_j)^2},$$

$$\hat{\mu}_j := \bar{W} - \hat{\beta}_j \bar{X}_j,$$

with $\bar{X}_j := \frac{1}{n} \sum_{i=1}^n X_{ij}$ and $\bar{W} := \frac{1}{n} \sum_{i=1}^n W_i$, and $\hat{j}$ maximises $|\hat{\rho}_j|$ over $j = 1, \dots, p$ with

$$\hat{\rho}_j := \frac{\sum_{i=1}^n (W_i - \bar{W})(X_{ij} - \bar{X}_j)}{(\sum_{i=1}^n (W_i - \bar{W})^2)^{1/2} (\sum_{i=1}^n (X_{ij} - \bar{X}_j)^2)^{1/2}}.$$

11. Consider the optimisation problem of performing a weighted empirical risk minimisation over the class of decision stumps. Specifically, suppose we have weights $w_1, \dots, w_n > 0$ and a single predictor whose observations have been sorted as $X_1 < \dots < X_n$. Show then that finding an ERM over

$$\mathcal{B} = \{x \mapsto \text{sgn}(x - a), x \mapsto \text{sgn}(a - x) : a \in \mathbb{R}\}$$

(i.e. minimising $\sum_{i=1}^n w_i \mathbb{1}_{\{h(X_i) \neq Y_i\}}$) may be performed in $O(n)$ computational operations.

12. In this question, we will study the Rademacher complexity of a simple neural network with a single hidden layer of m nodes, no biases, ReLU activation function ψ , and additional ℓ_2 -norm constraints on the parameters. Specifically, consider the set $\mathcal{H}$ of hypotheses of the form

$$h(x) = \sum_{j=1}^m \alpha_j \psi(\beta_j^\top x)$$

where $\alpha_j \in \mathbb{R}$ and $\beta_j \in \mathbb{R}^p$ for $j = 1, \dots, m$, with the constraints $\|\alpha\|_2 \leq \lambda_\alpha$ and $\max_{j=1, \dots, m} \|\beta_j\|_2 \leq \lambda_\beta$. Let $\mathcal{X} = \{x \in \mathbb{R}^p : \|x\|_2 \leq C\}$ and take $x_{1:n} \in \mathcal{X}^n$.

- (a) By considering $\mathcal{B} := \{x \mapsto \psi(b^\top x) : \|b\|_2 \leq \lambda_\beta\}$ and $\mathcal{B}' = \{x \mapsto b^\top x : \|b\|_2 \leq \lambda_\beta\}$, show that

$$\mathbb{E} \left(\sup_{b: \|b\|_2 \leq \lambda_\beta} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \psi(b^\top x_i) \right| \right) \leq \frac{2C\lambda_\beta}{\sqrt{n}},$$

where $\varepsilon_{1:n}$ are i.i.d. Rademacher random variables. [Hint: Apply the contraction lemma.]

- (b) Let us introduce the set of vector-valued functions $\mathcal{G} := \{g := (g_1, \dots, g_m) : g_j \in \mathcal{B} \text{ for } j = 1, \dots, m\}$. Show that

$$\hat{\mathcal{R}}(\mathcal{H}(x_{1:n})) = \lambda_\alpha \mathbb{E} \left(\sup_{g \in \mathcal{G}} \left\| \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(x_i) \right\|_2 \right).$$

- (c) Finally show that

$$\hat{\mathcal{R}}(\mathcal{H}(x_{1:n})) \leq 2C\lambda_\alpha\lambda_\beta \sqrt{\frac{m}{n}}.$$